

Notes: Van den Steen (AER, 2010)

Interpersonal Authority in a Theory of the Firm

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1 One-sentence summary

This note summarizes Van den Steen’s theory of the firm, which explains firms as institutional arrangements that use centralized asset ownership, low-powered incentives, and efficiency-wage discipline to generate *interpersonal authority* under open disagreement, thereby linking firm boundaries to decision externalities, differential confidence, and the motivation-cooperation trade-off.

2 Big picture

The paper starts from a basic organizational fact: inside firms, managers give orders and employees often obey, even when employees privately disagree. Standard property-rights theories define a firm as common asset ownership but do not directly explain why employees are more likely than contractors to obey. Van den Steen builds a model in which the firm itself is a mechanism for producing obedience.

Core idea. A firm is not only a collection of assets or contracts. It is also an arrangement that makes one person able to tell another what to do with a credible expectation of compliance. This authority emerges because asset ownership changes outside options, and fixed wages reduce the employee’s incentive to disobey when the employee disagrees with the manager.

The paper is theoretical. It studies two people, two projects, and two assets. Each project requires one asset and one decision. Decisions are noncontractible, payoffs are contractible, and people may have differing priors. In the market-like regime, each person owns one asset, executes one project, receives that project’s residual income, and has no authority over the other. In the firm-like regime, one person owns both assets, receives residual claims, pays the other a wage, and obtains interpersonal authority over the other.

3 Incremental contribution of the paper

The paper makes three main incremental contributions.

1. **It formalizes interpersonal authority as an equilibrium outcome.** Earlier theories often treat authority as primitive or define the firm through ownership. This paper derives authority from contracts, asset ownership, outside options, and disagreement.
2. **It explains why three common firm features appear together.** The theory jointly rationalizes centralized asset ownership, low-powered incentives, and managerial authority. These are not separate assumptions; they are mutually reinforcing equilibrium practices.
3. **It gives a new view of firm boundaries.** Integration is favored when cooperation and coordinated decisions are important; nonintegration is favored when motivation, effort incentives, or differential confidence about separate decisions are important. The paper also shows that better contractibility can lead to more integration, not necessarily less integration.

The contribution is especially important relative to Alchian and Demsetz’s critique that firms have no special authority beyond ordinary contracting. Van den Steen responds by showing why an employee may obey a boss more readily than a grocer obeys a customer: the employee typically does not own the productive asset, has low-powered incentives, and can lose an efficiency wage if fired.

4 Motivation and conceptual framework

4.1 What is interpersonal authority?

The paper defines interpersonal authority as a situation where one person tells another what to do, the second person tends to act according to the instruction, and the second person often does so against his own beliefs or immediate preferences. This is stronger than advice because the subordinate may obey even when he disagrees.

4.2 Why standard theories are incomplete

Property-rights theory studies ownership and residual control rights, but authority over people is not central. Relational contracting models can explain repeated cooperation, but they do not directly capture “abdication of choice.” Van den Steen wants a theory in which employees obey because the employment relationship itself changes incentives and outside options.

4.3 Why disagreement matters

The model assumes open disagreement: people may have different priors even when they share the same information. This is important because obedience is meaningful only when the subordinate might believe the boss is wrong. Under common priors, pay-for-performance can often align incentives. Under differing priors, however, high-powered incentives may make disobedience more likely because the agent believes his own action is better.

5 Model environment

5.1 Players, projects, assets, and decisions

There are two players, P_1 and P_2 , two projects, R_1 and R_2 , and two assets, a_1 and a_2 . Each project requires its corresponding asset and full-time involvement by one player. Each project also requires one decision:

$$D_k \in \{X_k, Y_k\}, \quad S_k \in \{X_k, Y_k\},$$

where S_k is the true state and decision D_k is correct if $D_k = S_k$.

5.2 Decision externalities

Let

$$d_k = \mathbf{1}\{D_k = S_k\}.$$

Project k succeeds with probability

$$Q_k = (1 - \theta)d_k + \theta d_{-k}, \quad \theta \in (0, 0.5).$$

The parameter θ captures decision externalities. When θ is low, each project’s outcome depends mostly on its own decision. When θ is high, coordinated decisions across projects are more valuable.

5.3 Beliefs and confidence

Each player has a subjective belief μ_k^i that state $S_k = X_k$. Players can disagree even without private information. Confidence is summarized by

$$v_k^i = \max\{\mu_k^i, 1 - \mu_k^i\}.$$

The paper focuses on confidence differences: the person who is more confident about a decision values control more highly.

5.4 Contracts and timing

Players negotiate asset ownership, project allocation, and payoff contracts. Decisions are not contractible, but project payoffs are contractible. Contracts combine a fixed wage w_k^i and a success-contingent share α_k^i :

$$\alpha_k^1 + \alpha_k^2 = 1.$$

Players can send cheap-talk messages, which become orders in the authority equilibrium. The subordinate then decides whether to obey.

6 Main mechanism

6.1 Efficiency wages create obedience

An employee may obey because disobedience risks firing and the loss of a wage. The wage is an efficiency wage: it is paid not just for labor input, but also to create compliance. If the wage is high enough, the employee obeys even when he thinks the boss's instruction is wrong.

6.2 Low-powered incentives make obedience easier

High-powered incentives make the employee care strongly about project success. When the employee disagrees with the boss, high-powered incentives increase the employee's temptation to follow his own belief. Low-powered incentives reduce this temptation. Therefore, fixed wages are not merely weak incentives; in this paper they are useful because they support authority.

6.3 Asset ownership strengthens authority

Asset ownership changes outside options. If the manager owns the asset, the manager can credibly threaten to fire a disobedient employee and continue with the asset. If the employee owns the asset, firing is less punitive because the employee keeps the asset or has a better outside option. Thus centralized asset ownership increases the manager's ability to enforce obedience.

6.4 Colocation of income and control

Under differing priors, the person with more control values residual income more, because that person believes her own decisions are better. Similarly, the person with more residual income values control more. Hence, income rights and control rights tend to be colocated. This logic pushes toward concentrated ownership and authority in the firm regime.

7 Proposition 1: integrated firm versus nonintegrated market

The central result partitions the parameter space into two regimes.

7.1 Firm-like integrated regime

In the firm-like regime:

- one player owns both assets;
- the asset owner is the full residual claimant on both projects;
- the other player receives a positive fixed wage and no residual income;
- the asset owner has interpersonal authority over the other player;
- the employee obeys because disobedience risks losing the wage.

This regime is favored when decision externalities are high, because coordinated decisions across projects are valuable.

7.2 Market-like nonintegrated regime

In the market-like regime:

- each player owns one asset;
- each player executes the project associated with her own asset;
- each player is the residual claimant on her own project;
- no player has authority over the other.

This regime is favored when decision externalities are low and when different players are most confident about different decisions.

7.3 Comparative static

As the decision externality θ increases, the equilibrium shifts from nonintegration to integration. Economically, higher θ means decisions are more interdependent, so centralized authority becomes more valuable.

8 Formal intuition: the obedience constraint

A key obedience condition can be written as a lower bound on the wage needed to make the subordinate obey. If P_2 executes project R_k and P_1 commits to quit if and only if P_2 disobeys, then P_2 obeys only if the wage exceeds a threshold depending on outside options, residual claims, disagreement, and decision externalities.

The economic meaning is simple:

- a higher employee outside option makes obedience harder;
- more residual income for the employee makes obedience harder under disagreement;
- asset ownership by the manager makes firing credible;
- low-powered incentives and positive wages make obedience easier.

9 Private benefits and moral hazard

9.1 Private benefits from actions

The paper shows that the main result is robust to moderate private benefits. If the employee gets a private benefit from taking a particular action, the efficiency wage must be high enough to make the employee forgo that private benefit. Authority still survives if private benefits are not too large.

9.2 Moral hazard extension

The model is extended to include unobservable effort. Effort is efficient but privately costly. Now pay-for-performance may be useful to induce effort. However, pay-for-performance also makes the employee more willing to disobey. This creates a core organization-design trade-off:

motivation through incentives versus cooperation through authority.

9.3 Three regimes in the moral hazard model

The moral hazard model has three parameter regions:

1. **Nonintegration:** no authority, each player owns one asset, both exert effort.
2. **Integration with full effort:** one player owns both assets, authority exists, and incentives are sufficient for both players to exert effort.
3. **Integration with partial effort:** authority exists, but only the manager exerts effort because giving the employee stronger incentives would weaken obedience too much.

10 Contractibility and the Coase-Knight distinction

A striking implication is that better contractibility does not necessarily reduce integration. When actions become contractible, some regions shift toward integration with full effort. This contrasts with a simple Coasian view in which firms substitute for incomplete contracts. In Van den Steen's Knightian interpretation, the firm is a mechanism for allocating control and residual income under uncertainty to the most confident decision maker. Therefore, better contracts can make the firm work better and can increase integration.

11 Empirical and organizational implications

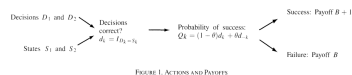
The theory predicts that integration is more likely when:

- uncertainty is high and disagreement is likely;
- decision externalities are important;
- cooperation is more important than individual motivation;
- important decisions require centralized direction;
- relation-specific investments would be harmed by future strategic disagreement.

It also predicts that people who own important assets are less likely to be subject to authority, and that general or specific human capital can weaken managerial authority by increasing employee outside options.

12 Figures

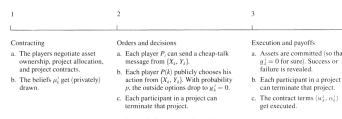
12.1 Figure 1: Actions and Payoffs; Figure 2: Time Line of Basic Model



Read: Each project has a binary decision and a binary state. Success probability depends mostly on the project's own correct decision and partly on the other project's decision.

Economic message: The parameter θ turns decisions into coordination problems; this is the primitive force for integration.

Figure 1. Actions and Payoffs

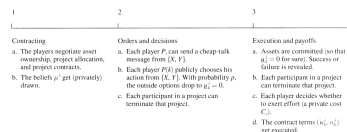


Read: The basic game moves from contracting to belief realization, orders, decisions, quitting opportunities, and final payoff execution.

Economic message: Authority is not assumed. It emerges because orders are cheap talk but disobedience can trigger firing and wage loss.

Figure 2. Time Line of Basic Model

12.2 Figure 3: Timeline of Model with Moral Hazard; Figure 4: The Tradeoff between Effort and Cooperation



Read: The moral hazard extension adds a post-success effort decision before contract payments are executed.

Economic message: Incentives are now needed for effort, but they can weaken authority by increasing the employee's temptation to disobey.

Figure 3. Timeline of Model with Moral Hazard

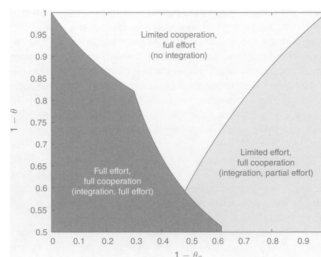


FIGURE 4. THE TRADEOFF BETWEEN EFFORT AND COOPERATION

Read: The parameter space is divided into nonintegration, integration with full effort, and integration with partial effort.

Economic message: Markets emphasize effort incentives; firms emphasize cooperation and authority. The optimal boundary depends on which margin matters more.

Figure 4. The Tradeoff between Effort and Cooperation

12.3 Figure 5: The Effect of Contractibility

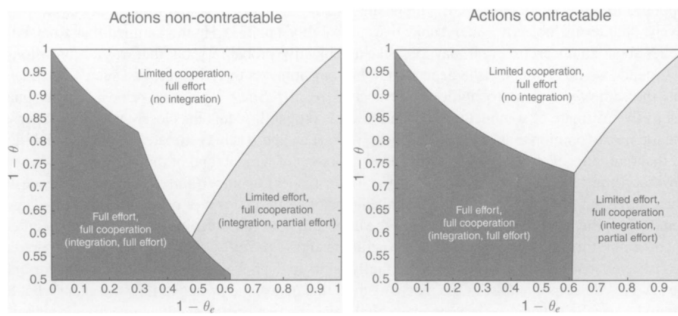


FIGURE 5. THE EFFECT OF CONTRACTIBILITY

Note: $\nu = 0.75$, $p = 0.1$, $B = \mu = 1$.

Read: The left panel shows the regime partition when actions are noncontractible; the right panel shows the partition when actions are contractible.

Economic message: Better contractibility can increase integration by improving contracting within the firm, not simply by replacing firms with markets. This is the paper's key Coase-versus-Knight point.

Figure 5. The Effect of Contractibility

13 How to read the model as a theory of the firm

13.1 Firm as authority-generating institution

The firm is a legal and organizational arrangement that aggregates asset ownership and contracts so that a manager can act as if she were the owner of the key assets and the counterparty to key contracts. This creates a basis for authority even in larger firms with dispersed shareholders.

13.2 Why employees obey more than contractors

Employees are more likely to obey because they usually do not own the relevant assets, receive low-powered incentives, and can be fired from the relationship. Contractors, by contrast, often own their assets and bear the consequences of their own decisions, so they are less subject to another party's authority.

13.3 Why fixed wages are productive

Fixed wages reduce disagreement-driven disobedience. In this theory, low-powered incentives are not a defect of employment contracts; they are part of how the firm creates cooperation under uncertain and contested decisions.

14 Limitations and caveats

- The model relies heavily on differing priors for the strongest results on low-powered incentives and colocation of income and control.
- The paper is theoretical and does not estimate the model empirically.
- The model focuses on two players, two assets, and two projects; large organizations require an extension through the legal-person interpretation of the firm.
- Authority is generated by firing and outside options; real organizations also use norms, identity, culture, career concerns, and relational contracts.

15 Short summary

Van den Steen develops a theory in which firms exist partly to create interpersonal authority. The mechanism is built from open disagreement, asset ownership, efficiency wages, and low-powered incentives. When decision externalities are high, centralized ownership and fixed wages allow one person to direct another and obtain compliance. When effort incentives or differential confidence are more important, nonintegration can dominate. The model therefore unifies asset ownership, low-powered incentives, and managerial authority as a coherent equilibrium bundle.

16 Conclusion

16.1 Core conclusion

The paper's central conclusion is that firms can be understood as mechanisms that give managers interpersonal authority over employees. Centralized asset ownership and low-powered incentives are not merely institutional details; they are necessary parts of the authority mechanism.

16.2 Incremental contribution revisited

The incremental contribution is to transform authority from a verbal description of firms into a formal equilibrium outcome. The paper connects classic Knight-Coase-Simon ideas to modern organizational economics and shows that the firm boundary is shaped by a trade-off between cooperation and motivation.

16.3 Takeaway

If a project requires coordinated decisions under uncertainty and disagreement, bringing the project inside the firm can give the manager the practical authority needed to coordinate actions. This is why ownership, wages, and hierarchy appear together in firms.